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Malaysia: Reducing Food Dependency for a More Sustainable Population

Background

Malaysia is located in Southeast Asia and has an estimated population of 35 million, with about 78% living in urban areas and 22% of the population in rural areas (CIA, 2024). It runs under a parliamentary constitutional monarchy (CIA, 2024) and scores 41st on the global food security index (Economist, 2022). While this ranking may seem high, about 25% of the population—nearly 8.7 million people—has experienced food insecurity (Mahmood et al., 2022). Currently, only 26.1% of the land is cultivated (CIA, 2024), with the average farm being roughly 2 to 8 hectares (Oliveira, 2013), (3 to 12 soccer fields) (Metric Views, 2007). Major agricultural products include rice, chicken, eggs, and tropical fruits like coconuts, pineapples, and bananas. Their main exports are rubber and palm oil. Malaysia is a tropical country with coastal plains, hills, and mountains that form the landscape (CIA, 2024).

The typical family size is two parents and two children living in a small village outside a regional town. The average household median income is about RM 7017 or \$1778 USD a month (DOSM 2024), and the most popular job sectors are services (51.9%) and manufacturing (27.7%), with agriculture only accounting for 5.5% of employment despite having 16.5% of the job vacancies (Lockard & Ooi Jin Bee, 2019). The parents may operate a 2 to 8-hectare farm to grow rice, durian, or pineapples, and other local tropical fruits and vegetables like papaya, guava, and cabbage. They may supplement their diet with foods like poultry, noodles, and fish (like mackerel and snapper) (DOSM, 2020-2024). Occasionally, the family might go to wet markets, grocery stores, or supermarkets to get any additional items (Chamhuri et al., 2013; Krin Institute, 2024). They can cook their food using gas or electric stoves (New Malaysian Kitchen). The family has access to affordable healthcare services and high-quality, affordable education (International Trade Administration, 2023; WHO, 2024; Wahaj, 2025). Moreover, they have access to special government programs, including welfare relief systems for senior citizens, assistance for lower-income individuals, and special programs for the disabled. (Lockard & Ooi Jin Bee, 2019). They have extensive access to infrastructure, as shown in the following figure:



(Ekonomi, 2011-2015)

While this family can grow their own healthy food and is supported by Malaysia's robust systems, they still face food problems. How could a country with such strong development still face food security challenges? It is due to Malaysia's increasing reliance on cheap, highly processed imports (DOSM, 2020-2024). This makes day-to-day life difficult for everyday rural families.

While Malaysia has extensive access to resources, it's still not food independent. At a local level, general trends show that urban families are more and more likely to eat out over time rather than cook food by themselves. Data shows that food expenditures have increased from $\frac{1}{3}$ of family total food expenditures to nearly $\frac{1}{2}$ (Krin Institute, 2024). Evidently, at a local level, Malaysian residents are increasingly dependent on restaurants to make their food because it is convenient (Krin Institute, 2024). However, this food dependency has led to the widespread use of online food ordering platforms. (Krin Institute, 2024). While consumers enjoy greater convenience and wider choices when buying food online, they are exposed to increasingly unhealthy options. Because the food service industry prioritizes profit margins, they rely heavily on cheap, highly processed ingredients imported from abroad rather than sourcing from local Malaysian farmers. Therefore, the domestic shift toward eating out actively fuels Malaysia's dangerous reliance on international food imports, driving up both the import rate and the cost of a healthy, locally sourced diet (Up 26%). (Krin Institute, 2024).

This actively promotes unhealthy food consumption by reducing the emphasis on nutritious fruits and vegetables (Krin Institute, 2024). This leads us to the central issue: Malaysia's lack of sustainable agriculture. Left unaddressed, this problem spirals into malnutrition, food economy volatility, and a lack of innovation and interest in the agricultural sector.

Problem

Currently, the effects of Malaysia's food dependency are evidenced by both the Ukraine War and COVID-19. It's established that Malaysia's food security was significantly impacted by these global events. (Nur et al., 2025; Wong et al., 2022). Additionally, their own food economy is affected by their own weather. Hot temperatures, floods, rising sea levels, and environmental pollution further hinder Malaysia's food security. (Lockard & Ooi Jin Bee, 2019; Nur et al., 2025; Mahmood et al., 2022; Ramli et al., 2023). Malaysia only produces about 31% of the caloric requirements locally, and currently, the government is giving the agriculture sector lots of subsidies, yet it is still struggling (Devaraj, 2025). Trends have shown that the average spending on food imports is increasing at an alarming rate (Dardak, 2025). Moreover, young people see a disconnect between future careers and agriculture. Agriculture has little to no appeal to them (Nur et al., 2025; Ministry of Agriculture and Food Industries Malaysia, 2021). This results in the problem of aging farmers. Elderly people are carrying the burden of farming production with nobody to replace them. The Gross Domestic Product (GDP) for agriculture has shrunk from $\frac{1}{3}$ of the GDP to $\frac{1}{10}$. Similar trends can be seen in labor force engagement for agriculture. (Lockard & Ooi Jin Bee, 2019).

The main factors contributing to this problem are land misallocation (a large majority of land is dedicated to cash crops like oil palm) (Devaraj, 2025; Wong et al., 2022), Malaysia's volatile climate, which reduces domestic crop yields (Mahmood et al., 2022; Nur et al., 2025; Ramli et al., 2023), and low levels of Research and Development (R & D) (Dardak, 2025; Nur et al., 2025; Ramli et al., 2023).

As a result, rural and lower-income individuals have difficulty getting access to food (Dardak, 2025), and the entire population is impacted by increased risks of cardiovascular diseases and diabetes. This contributes to less agricultural productivity (Mahmood et al., 2022), increases social disparities (Nur et al., 2025), and contributes to global warming (Mahmood et al., 2022; Ramli et al., 2023; Ministry of Agriculture and Food Industries Malaysia, 2021). The largest impact can be seen on the rural population; rural farmers can't compete with the imported goods. This, in combination with the extremely volatile climate, has resulted in farmland being converted for other uses (Devaraj, 2025). Moreover, farming has little to no appeal to the youth, leaving rural populations in shambles. This problem impacts urban populations through health risks, rural communities that struggle to sustain their careers, and low-income individuals for whom food price volatility can mean the difference between nourishment and starvation.

Solution Inspiration

To address this food dependency, the following solutions are most popular for other countries facing similar problems: technological modernization and youth empowerment, strategic land reallocation, and increased nutritional awareness. These solutions would give more jobs for the domestic economy, reduce the price volatility of food crops, increase youth interest in the declining agricultural sector, and lead to a healthier population. While it doesn't directly address pollution, it is assumed that the R & D would lead to higher agricultural efficiency and lower emissions. By establishing self-dependency and modernizing, Malaysia can produce food more effectively and develop resilience to price volatility.

Malaysia's local government has already enacted some of these solutions through their National Agricultural Policies, which increase R & D, increase agricultural productivity, development sustainability, and food security (Nur et al., 2025). The government is aware of the problem and is directly addressing it. However, the enacted solutions have their limits: the solutions are non-unified (low collaboration amongst different sectors), there is no focus on which foods to produce, industrial crops are prioritized, price ceilings are focused on unhealthy foods, and they don't address food waste (Wong et al., 2022).

To address these vulnerabilities, Malaysia can also draw inspiration from neighboring nations: China's integration of e-commerce in agriculture (through Alibaba) and Singapore's heavy investment in public-private agricultural R & D and diversified imports (Wong et al., 2022) are examples of solutions other nations have implemented to address their food dependency problems.

Final Solution

To address this problem, this proposal uses the OGSM (Objectives, Goals, Strategies, and Measures) model. With the clear objective of establishing 70% food self-sufficiency by 2030 for the production of meat, vegetables, and carbohydrates (modeled off of IMANI, 2025; Mohamed et al., 2025), the chosen strategy was to invest more in R & D and place more emphasis on agricultural land rather than maximizing profits with cash crops.

Increased agricultural investments would support domestic agribusinesses and agricultural technology. It implies investing in technology and selecting the right crops to grow using the Total Factor Productivity Index (Ramli et al. 2023), and using this index to pinpoint and focus on efficient and inefficient production items. Moreover, this also means increasing public-private partnerships to help fund these new agricultural innovations. This strongly mirrors Singapore's framework, where a large private company, Temasek (Wong et al., 2022), collaborated with the government to provide \$750 million for agricultural investments. It took as little as two months to secure the funds and start investing (Limited, 2013).

Diving into the different partnerships, the Ministry of Agriculture and Food Security (drives domestic production), the Ministry of Domestic Trade and Cost of Living (manages price controls), and the Ministry of Health (ensures food safety and nutritional standards) must collaborate with large agribusinesses, agricultural tech startups, and local financial institutions to provide essential loans to small-scale producers. Furthermore, the local farmers, youth entering the agricultural workforce, and everyday consumers can provide critical ground-level feedback and drive market demand. Malaysia must also implement targeted subsidies and agri-tech tax laws in which the government redirects existing agricultural subsidies away from cash crops and exclusively toward healthy staple crops. Legal tax holidays (such as 10-year exemptions) can be enacted for private companies and foreign investors that fund domestic agricultural R&D and technology (Wong et al., 2022; IMANI, 2025).

Additionally, Malaysia can increase the percentage of land legally dedicated to domestic agriculture by 20% by utilizing protective zoning and reclamation strategies (Weber & Buckingham, 2016). To achieve this part of the solution, the Malaysian government must implement the following legal and economic frameworks: the Youth Land Rights and Agricultural Zoning Protection Acts. The former is implemented to circumvent the high cost of land; the government implements a state-sponsored leasing policy (20-30 year leases) exclusively to young "Agropreneurs" who commit to growing staple food crops using modern technology (inspired by Mauto, 2020). The latter enforces strict zoning laws that are closed off as "Permanent Food Production Zones." This will legally block private corporations from buying designated food-crop land and converting it back into lucrative palm oil or rubber plantations (inspired by Corcoran, 2025). This land-use conversion process could take approximately 6-12 months (Land, 2024).

The costs associated with this are not extreme. The following table shows the average GDP of various sectors of the economy.

Metric (2021 DOSM Data)	Agriculture	Manufacturing	Services
Gross Output	RM 101.3 Billion	RM 1,609.3 Billion	RM 1,174.5 Billion
Value Added (GDP)	RM 55.7 Billion	RM 355.1 Billion	RM 658.2 Billion

(DOSM, 2021)

An exemplar nation that Malaysia can draw from is Qatar. They have heavily invested in high-tech agriculture and converted a large portion of their land with an emphasis on agriculture and food manufacturing, with just a \$1-1.5 billion (RM 5.9 billion) agricultural investment (IMANI, 2025; Mohamed et al., 2025). As a result, they achieved full self-sufficiency in dairy and livestock production, exceeded 100% self-sufficiency in fresh poultry production, and increased vegetable production by 36% over the span of five years (Mohamed et al., 2025).

With Malaysia's strong economy, similar results can be achieved. Malaysia has extensive mineral resources, and they're mass producers of gas, palm oil, rubber, and electronic equipment (Lockard & Ooi Jin Bee, 2019). Malaysia has a great economy and a government that cares, so it's extremely feasible for them to solve this problem. Evidently, a significant portion of funds generated from manufacturing and the services sector can be invested in the agriculture sector. Currently, Malaysia has a 2026 budget of RM 2.62 billion (\$667 million- nearly ½ of the investment required to bolster Qatar's self-sufficiency) to support rice farmers through price support, crop input, fertilizer, seed, and production incentives, the highest agricultural subsidy in Malaysia's history (Malaysian Reserve, 2026). Supplementing this subsidy with subsidy reallocations from the manufacturing and services sectors would significantly improve the gross output of the agricultural sector. In order to achieve this, the Malaysian government would need to pass a modified supply bill to redirect these funds to the agriculture sector (Parliament of Malaysia, 2026). To gain support for the bill, the government could use mainstream media to raise public awareness about the seriousness of the food issue, provide incentives (such as tax credits and other benefits) for purchasing local products, and strongly promote healthy eating habits.

To ensure the effectiveness of these interventions, the Malaysian government and partnering organizations need to incorporate the following implementation strategies and metrics to ensure the success of this solution. These metrics can include:

- Self-Sufficiency Ratio (SSR) (modeled after QNA, 2022): Tracked annually by the Ministry of Agriculture and Food Security, calculating the annual production percentages specifically for targeted healthy staple crops (using the Total Factor Productivity Index to determine production inefficiencies) (Ramli et al. 2023)
- Import Dependency Ratio (IDR) (modeled after Global Economy, 2024): Calculated by the Ministry of Domestic Trade, tracking the ongoing dependency percentage using export revenue, domestic production, and import spending data.
- Global Food Security Index (GFSI): Monitored by the Ministry of Health to track Malaysia's annual movement within the global rankings.
- Youth Agricultural Engagement (modeled after IFAD, 2024): Track the number of new agribusiness registrations by founders under the age of 35, alongside enrollment rates in agriculture-related academic fields.

- **Sector Investments:** Calculate the annual volume of both Foreign Direct Investment (FDI) and Domestic Direct Investment (DDI) directed toward agribusiness and agricultural technology.
- **Land Utilization:** Estimate the total hectares of active food-farming land and track the conversion rate of cash crop plantations back into domestic food production.

The solution can be implemented in a five-year timeline:

- **Phase 1 (Year 1): Legislative Action.** The federal government passes the Agricultural Zoning Protection Act, legally gazettement the 20% land target to bypass the standard 6-12 month conversion delay. Simultaneously, initial R & D funds are injected into solar-powered agricultural tech hubs.
- **Phase 2 (Years 2-3): Pilot Rollout.** The state-sponsored 20-30-year land leases are awarded to the first wave of youth Agropreneurs. Smart indoor farming hubs and localized, tech-driven staple crop production begin.
- **Phase 3 (Years 4-5): National Scaling.** Successful pilot farms are scaled nationwide to fully hit the 20% land dedication target, systematically lowering the IDR and securing domestic supply chains against global shocks.

While this solution directly addresses food security and is beneficial for the population's health, this solution could benefit from the inclusion of women in the plan and more consideration of community behavior and attitudes. It's difficult to shift a cash crop prioritization to food sustainability, because cash crops are extremely profitable for Malaysia, but finding a way to do so will benefit Malaysia's health and improve its economic sustainability in the long term. Additionally, finding a way to get women involved would increase the labor force participation rate while giving women more job opportunities.

Cultural attitudes must be considered when implementing this solution. Currently, cultural pressure from parents always pushes youth towards engineering, medical, and corporate jobs (Monash University, 2024) instead of pursuing farming work; it's often seen as dirty, dangerous, and difficult work (Alifah Zainuddin, 2025). To make this sustainable agriculture plan work, the government must reshape public perceptions about farming.

Through this solution, sustainability is strongly promoted. The youth would be drawn to farming due to increased R & D, there would be improved food security, and innovation would flourish in the agricultural sector. This would also make farming cost-efficient, land-efficient, and environmentally friendly long-term, and reduce the social disparities between low-income individuals, rural, and urban communities. By creating a healthier population, we are increasing job opportunities while mobilizing more of the labor force. People live longer and healthier, which increases economic productivity and growth, which means more foreign investments, more opportunities for future growth, more educational opportunities for the youth, and more mobilization of lower-income individuals. Ultimately, this solution aims to reduce obesity, build a self-sustaining and united Malaysia across all sectors, and revive the declining agricultural economy to create more jobs for future generations.

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